Effat University
College of Engineering
Department of Computer Science

جامعة عفت
EFFAT UNIVERSITY

CS 3081 5 Oct 2026

CASSIA

Workload, Energy, Heat & Water Optimization

Application: AI Data Center Supporting Hajj Services

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THE PROBLEM

AI data centers consume large amounts of energy and water to operate and cool their servers. In Saudi Arabia, producing cooling water through desalination also requires significant energy. During Hajj, AI services may experience high and changing workloads, which can increase the demand for both energy and cooling.

OUR SOLUTION

01



Solar Energy

02



AI Workloads

03



Heat Recovery

04



Desalination

05



Water Storage

06

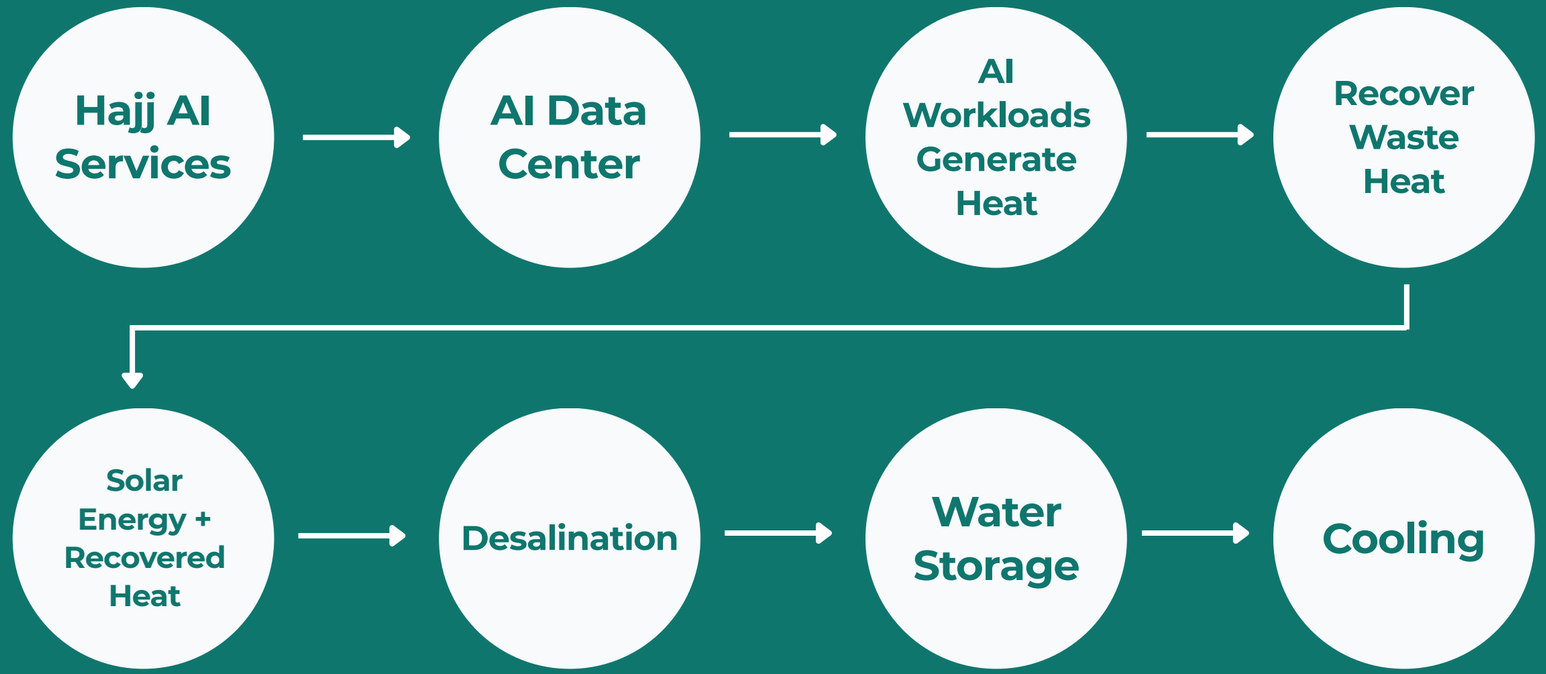


Cooling

We propose a simulated AI data center that combines solar energy, workload management, heat recovery, desalination, and water storage. An intelligent A* agent decides how to use these resources efficiently while keeping critical Hajj AI services running.



HOW IT WORKS


Hajj AI Services
AI Data Center
AI Workloads Generate Heat
Recover Waste Heat
Solar Energy + Recovered Heat
Desalination
Water Storage
Cooling

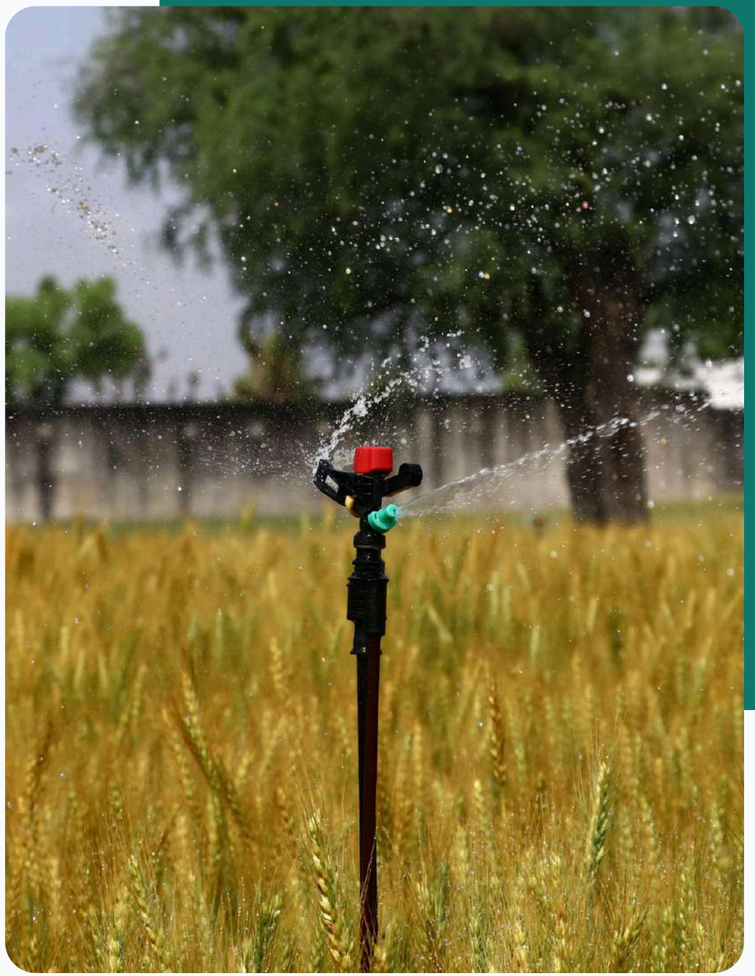
AI TECHNIQUES

A* Search
Used to find an efficient sequence of actions for managing:

- AI workloads
- Energy
- Heat recovery
- Water production
- Cooling

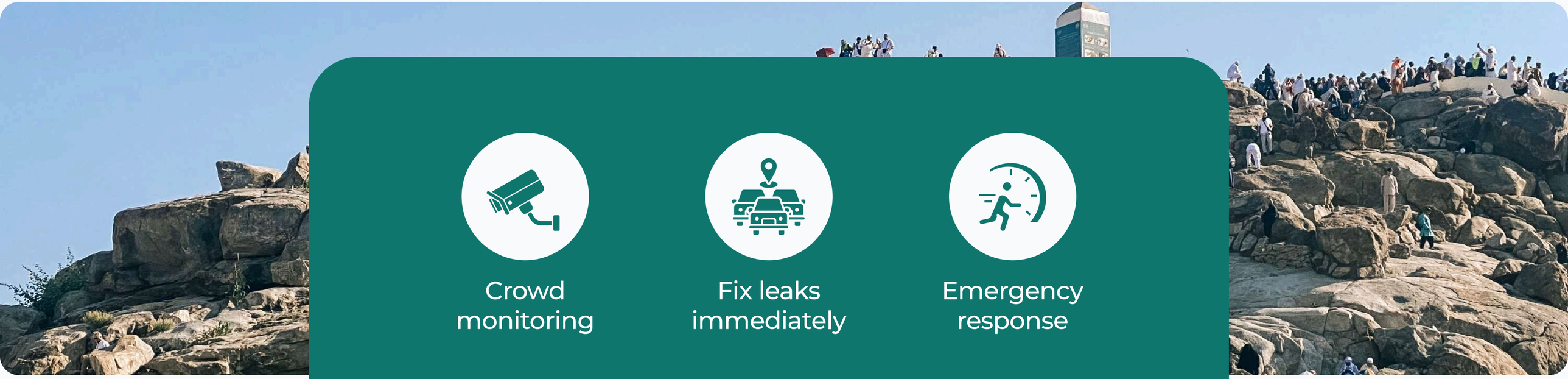
Optimization
The agent aims to:

Reduce external water and energy use while keeping critical AI services running.



HAJJ SCENARIO

The data center can support AI services such as:



Crowd monitoring

Fix leaks immediately

Emergency response

Some workloads are critical, while others can be delayed. This allows the agent to prioritize important services during periods of high demand.

CHALLENGES

The main challenge is balancing AI workload, energy, heat, and water. Solar energy and AI workloads can change over time, while the data center must still maintain enough cooling and keep critical Hajj services running. The system must make efficient decisions without wasting resources.

01



Managing limited water resources

02



Balancing energy and water use

03



Handling changing AI workloads

04



Maintaining cooling requirements

05



Keeping critical services running

WHY IT MATTERS

The project aims to reduce the data center's dependence on external freshwater and grid energy. It also demonstrates how waste heat and renewable energy can be reused to support cooling, making AI data centers more sustainable.

EVOLUTION

Initial Idea
Water-Self-Sufficient AI Data Center

Added Resource Management
Workload + Energy + Water

Added Heat Recovery
Reuse server heat

Added Hajj Scenario
Apply the system to Hajj-related AI services

Final Idea
An A-based intelligent system* that manages workloads, energy, heat, and water to improve data center sustainability.





MAIN GOAL

Reduce external freshwater and energy use while maintaining the performance of critical AI services.

- Solar Energy
- AI Workloads
- Heat Recovery
- Desalination
- Water Storage + A Search*

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THANK YOU

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